THE TOP COMPACT COHOMOLOGY OF THE LISSE SHEAF ATTACHED TO A REPRESENTATION IS THE COINVARIANTS WITH TATE TWIST

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For an irreducible curve C over kbar and a constant sheaf M, H_c^2(C, M) = M. invoked at proof line 3467. Provides the top cohomology of the constant sheaf on the components of the cover.
- H_c^2(X_kbar, mu_n) is canonically Z/nZ. invoked at proof line 3492. Explains the Tate twist in the Galois action on the identified coinvariant module.

Lemma 0.2. *There is a canonical isomorphism*

$$H^2_c(X_{\overline{k}}, \mathcal{F}_\rho) = (M)_{\pi_1(X_{\overline{k}}, \overline{\eta})}(-1)$$

as $Gal(k^{sep}/k)$*-modules.*

Proof of Lemma 0.2. Let $Y \to^{\varphi} X$ be the finite étale Galois covering corresponding to $\mathrm{Ker}(\rho) \subset \pi_1(X, \overline{\eta})$. So

$$\mathrm{Aut}(Y/X) = Ind(\rho)$$

is Galois group. Then $\varphi^*\mathcal{F}_\rho = \underline{M}_Y$ and

$$\varphi_*\varphi^*\mathcal{F}_\rho \to \mathcal{F}_\rho$$

which gives

$$\begin{aligned} & H^2_c(X_{\overline{k}}, \varphi_*\varphi^*\mathcal{F}_\rho) \to H^2_c(X_{\overline{k}}, \mathcal{F}_\rho) \\ & = H^2_c(Y_{\overline{k}}, \varphi^*\mathcal{F}_\rho) \\ & = H^2_c(Y_{\overline{k}}, \underline{M}) = \oplus_{\substack{\text{irred. comp. of} \\ Y_{\overline{k}}}} M \end{aligned}$$

$$\mathrm{Im}(\rho) \to H^2_c(Y_{\overline{k}}, \underline{M}) = \oplus_{\substack{\text{irred. comp. of} \\ Y_{\overline{k}}}} M \to_{\mathrm{Im}(\rho)\text{equivalent}} H^2_c(X_{\overline{k}}, \mathcal{F}_\rho) \to^{\substack{\text{trivial Im}(\rho) \\ \text{action}}}$$

irreducible curve $C/\overline{k}$, $H^2_c(C, \underline{M}) = M$.

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Since

$$\begin{matrix}\text{set of irreducible} \\ \text{components of } Y_k\end{matrix} = \frac{Im(\rho)}{Im(\rho|_{\pi_1(X_{\overline{k}},\overline{\eta})})}$$

We conclude that $H^2_c(X_{\overline{k}}, \mathcal{F}_\rho)$ is a quotient of $M_{\pi_1(X_{\overline{k}},\overline{\eta})}$. On the other hand, there is a surjection

$$\mathcal{F}_\rho \to \mathcal{F}'' = \begin{matrix}\text{sheaf on } X \text{ associated to} \\ (M)_{\pi_1(X_{\overline{k}},\overline{\eta})} \leftarrow \pi_1(X, \overline{\eta})\end{matrix}$$

$$H^2_c(X_{\overline{k}}, \mathcal{F}_\rho) \to M_{\pi_1(X_{\overline{k}},\overline{\eta})}$$

The twist in Galois action comes from the fact that $H^2_c(X_{\overline{k}}, \mu_n) =^{\text{can}} \mathbf{Z}/n\mathbf{Z}$. $\square$